

MoveScan

A moving estimate from your phone, in minutes.

// photograph each room → AI inventory → priced move → a human confirms

The survey is the bottleneck.

Every quote for a full-service move starts with someone driving out to walk the house with a clipboard. It is slow, it has to be scheduled, and it **caps how many leads a mover can quote in a week**.

Customers wait days for a number. Movers lose the ones who booked with whoever answered first.

TODAY'S SURVEY

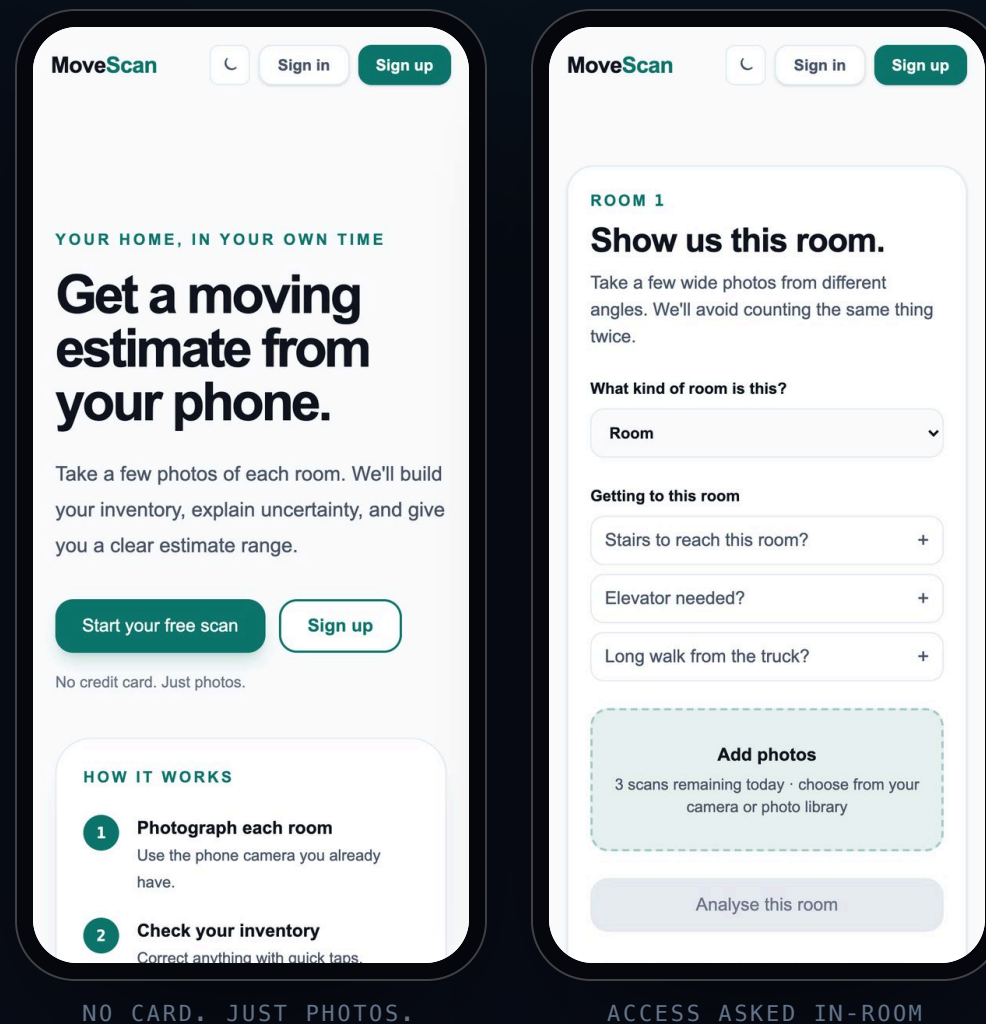
1 visit

PER ESTIMATE · SCHEDULED · IN PERSON

Phone-first, and calm about it.

The customer starts a scan with no account and no card. Each room asks the two things a mover needs on site — **stairs, elevator, long carry** — then takes a few wide photos.

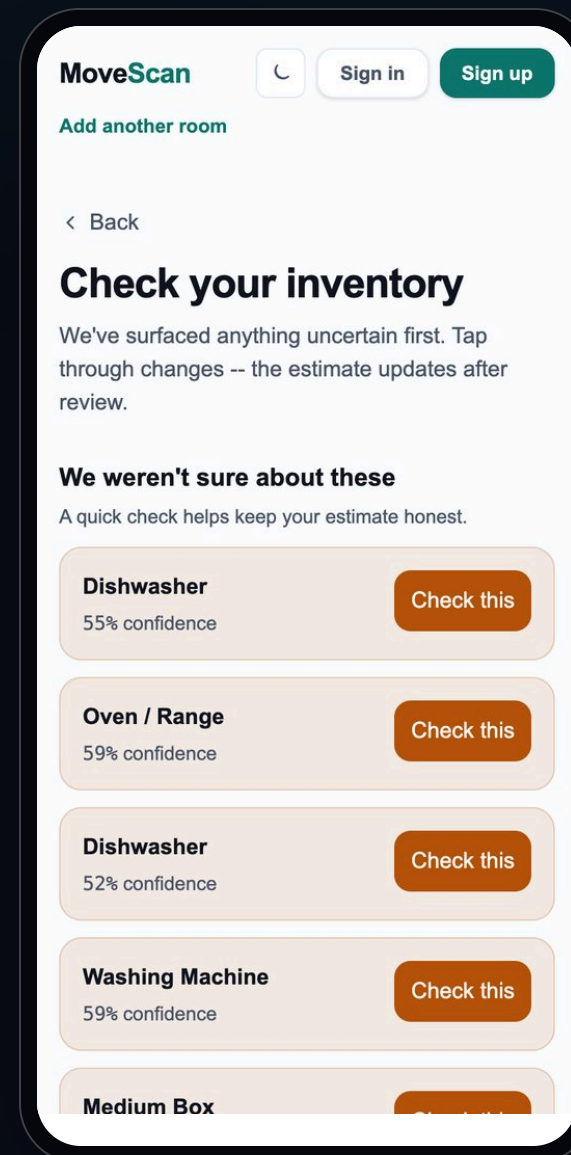
Loading copy narrates the real work: "Checking for duplicates across your photos."



It tells you what it's **not sure about** — first.

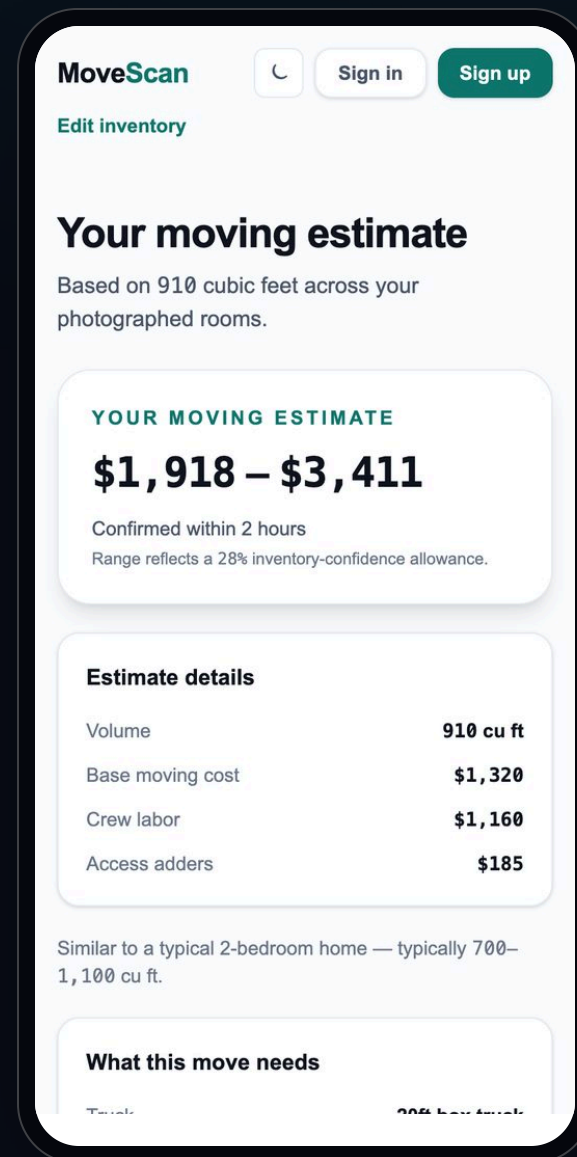
Every AI demo this year says "the model sees a photo and names things." Judges have seen it. The interesting part is the other direction.

- **Low-confidence items surface at the top**, with the confidence % and the model's own stated reason.
- **"Partly hidden behind the sofa."** **"Only visible from one angle."** — the model says why, in plain words.
- One tap re-checks a single item against the catalogue.



A range, not a fake-precise number.

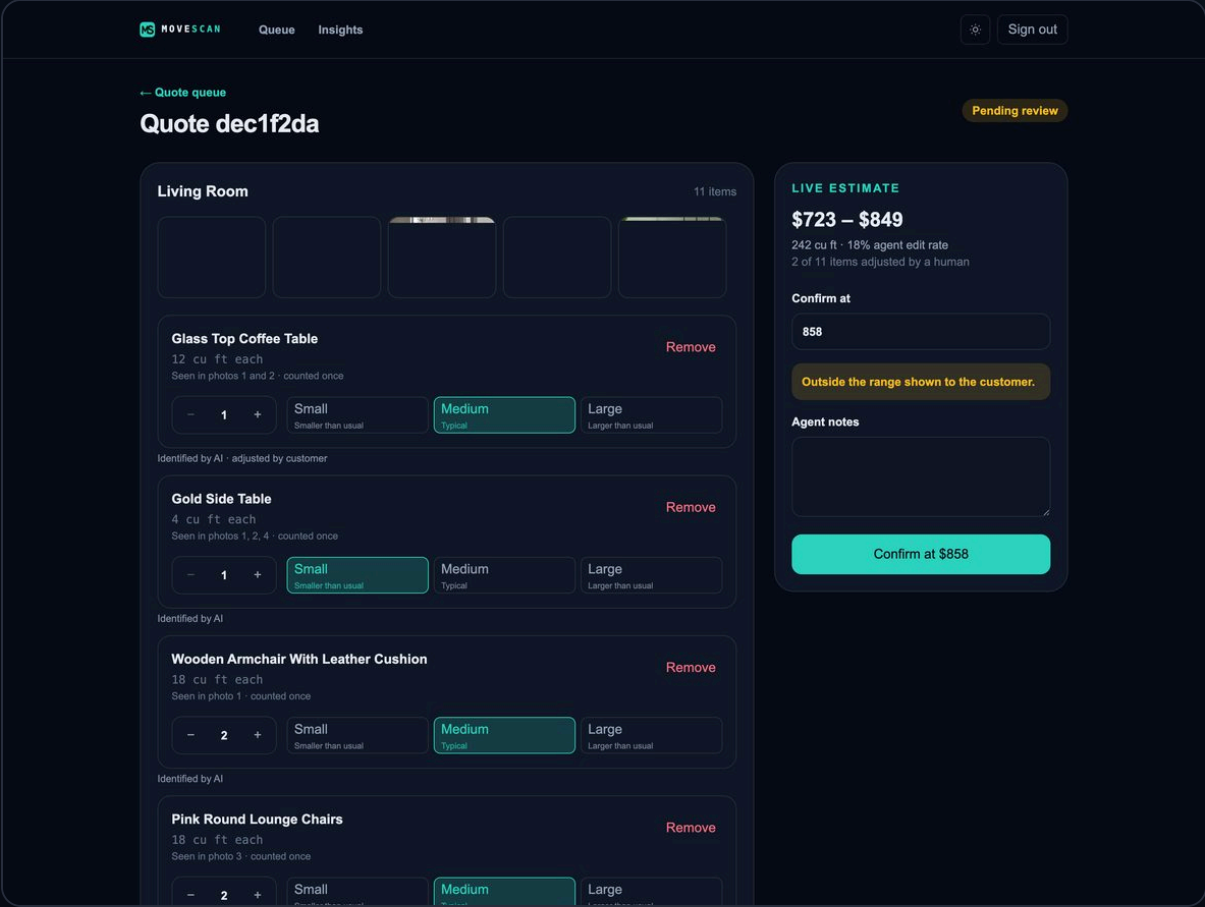
- The customer sees a **range**. It **widens** when the inventory is uncertain — computeTolerance adds a band for every low-confidence or user-added item.
- A human specialist **narrows** it to a firm price. It never jumps to a different number — nothing reads as bait-and-switch.
- The console reports **inventory accuracy as a measured figure**: the share of AI items a human accepted unchanged. An observation, not a claim.



Three photos of one sofa is one sofa.

All of a room's photos go into a **single Gemini call**, so the model can reason about object identity across frames instead of counting each photo separately.

Each item carries `seenInImages`, so the review and the dispatcher both show the evidence: "seen in photos 1 and 2 · counted once."



The inventory becomes a moving plan.

Truck class, crew size, hours, packing materials and special-handling flags are all **derived from the inventory** — deterministically, in `lib/moving-plan.ts`, with no extra AI calls.

Pure functions. Unit tested. "20ft box truck · 3 movers · 11 hours" is arithmetic, not a guess.

Base moving cost

\$1,320

Crew labor

\$1,160

Access adders

\$185

Similar to a typical 2-bedroom home — typically 700–1,100 cu ft.

What this move needs

Truck

20ft box truck

Crew

3 movers

Estimated time

11 hours

Special handling

Heavy

Dishwasher, Oven / range, Freezer, Washing machine, Upright piano, Grand piano

Fragile

China cabinet, Framed artwork

High value

China cabinet, Upright piano, Grand piano, Framed artwork

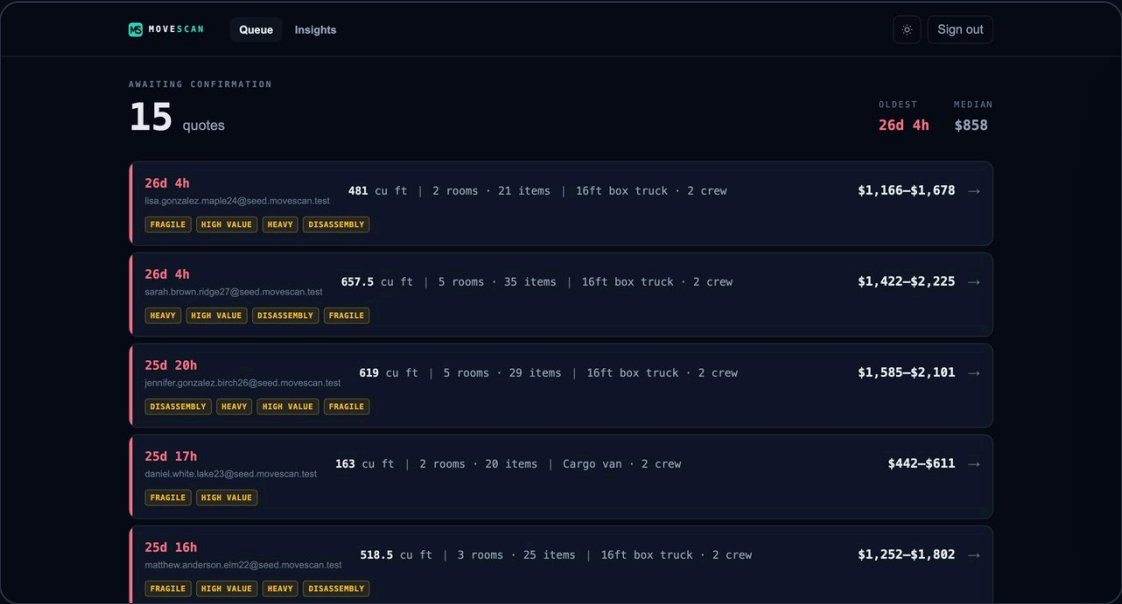
Disassembly required

Dining table, Upright piano, Grand piano

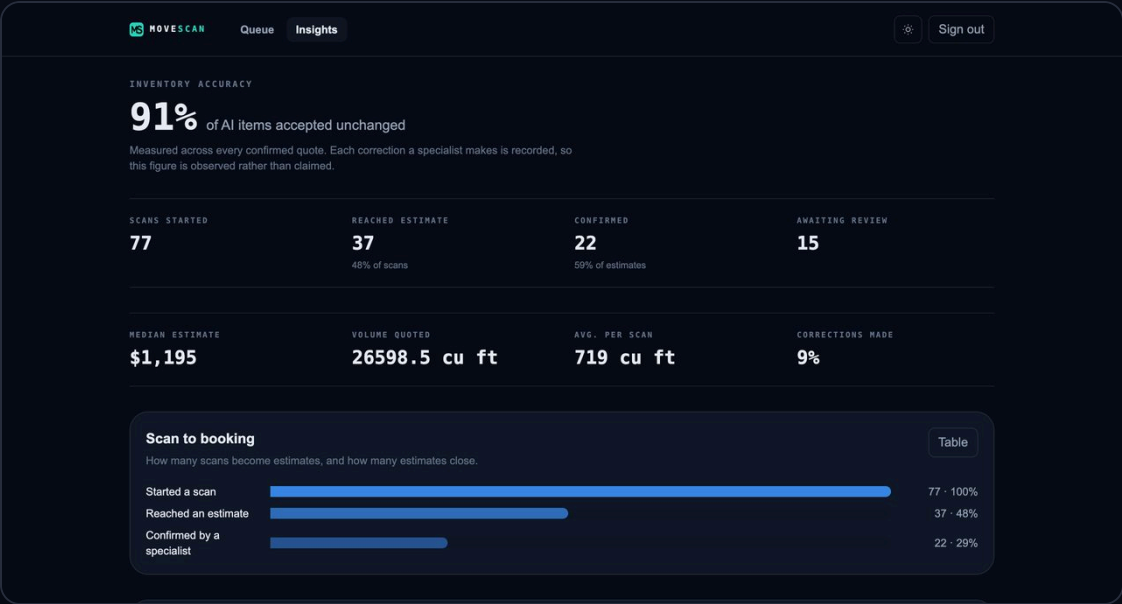
► Packing materials · 7 items · tap to see the list

A different user, so a different console.

The customer flow is light and spacious. The dispatcher works a queue under time pressure — so it's a dark ops console: urgency rails, handling chips, mono ticket rows, and an accuracy hero computed from real edits.



Queue — sorted by how long a customer has waited



Insights — accuracy, funnel, trend, composition

One AI box. Everything after it is deterministic.



■ AI – CLASSIFIES ■ DETERMINISTIC – COMPUTES & PRICED

Judgement, not just features.

The model classifies, the code computes

`pricing.ts`, `moving-plan.ts`, `catalogue.ts`, `chart.ts` import no network, no DB, no `process.env` — so every arithmetic step is unit-tested without spending API quota.

A model cascade that cannot throw

Classifies the error first: malformed JSON → retry the same model; a 429 → escalate to a model with its own quota bucket; both fail → checked-in fixtures behind a banner. Never an error page.

Authorization in the app layer, on purpose

Every query uses the service key, which bypasses RLS by design. `session-access.ts` guards the customer routes; "not yours" and "not found" both return **404** — a 403 would confirm the id exists.

Rate limiting is one atomic Postgres function

`consume_quota` locks every bucket row before checking, then spends from all of them or none — concurrent scans can't both take the last unit. Per-device, per-IP, per-account.

INFRASTRUCTURE SPEND

\$0

NO CARD ON FILE ANYWHERE

Free tier, end to end.

- **Vercel** Hobby — Next.js hosting
- **Supabase** free tier — Postgres + private Storage bucket
- **Gemini** free tier — Flash-Lite primary, chosen for quota headroom; reversible in one env var

Honest about the edges.

- **Recognition accuracy on real photos is not formally measured.**

The console's "accepted unchanged" figure is computed from seeded demo data — a demonstration of the mechanism, not a validated model claim.

- **Photo PII.** Interior photos catch faces and documents. Currently demo-grade: private bucket, signed 1-hour URLs. Production needs retention limits and deletion-on-request.

- **Currency.** Pricing is hard-coded USD; localisation is pending.

- **Regulated estimates.** US interstate moves fall under FMCSA rules (49 CFR 375). Agent review is the compliance control; a real price needs legal review first.

- **One rate card.** Real movers price per cu ft, per lb, hourly or from tariff tables. A general pricing engine is its own project.

- **Travel fare** from pickup/destination distance isn't built yet — see next slide.

Travel fare, by map pin.

The next feature adds distance-based travel cost from pickup and destination. The spec calls for a **map pin-drop, not typed addresses** — street addresses don't resolve reliably in Pakistan, where this is aimed.

That reasoning — design for where the users actually are — is the point.

VERIFIED AT DEMO BUILD

131 / 13 **tests** passing across 13 files

0 **typecheck** errors · **build** compiles

58 catalogue **items** · pure-function pricing

~7,900 **lines** across 96 files · 79 commits since Aug 31

91% All items **accepted unchanged** (seeded data)

200 **live** · smart-logistics-taupe.vercel.app